AADHAAR
The Greatest Testimony to FOSS' Success in India!

It is heartening to see how FOSS is powering one of India's most prestigious and mammoth e-governance initiatives—the *Aadhaar* project from the Unique Identification Authority of India.

For those of you who need an introduction to the *Aadhaar* project, it is a 12-digit individual identification number issued by the Unique Identification Authority of India (UIDAI) on behalf of the Government of India. This number will serve as a proof of identity and address, anywhere in India. Each *Aadhaar* number will be unique to an individual and will remain valid for life. By providing a clear proof of identity, *Aadhaar* is targeted at empowering the poor and underprivileged residents of the country to access services such as the formal banking system, and will give them the opportunity to easily avail various other services provided by the government and the private sector.

More than 6 crore UIDs have been issued till date, in an 18 month period. The UIDAI systems were recently scaled to process up to 1 million UID applications per day. Enrolments are actively happening in many Indian states. (Refer to http://portal.uidai.gov.in for daily updates.)

The project is historic since it is one of the largest e-governance initiatives in India till date and the scale is unprecedented. What makes it all the more interesting is the fact that the project is driven extensively by the FOSS stack.

Where FOSS is the first and foremost choice

The UIDAI technology team had indicated its preference for open standards and open source, says Regunath Balasubramanian, principal architect, the UIDAI project. "We were prudent in the very early stages of the project. That has continued to be the project's guiding philosophy. Initially, only the enrolment client application had a dependency on device drivers of a certain operating system, and that too was for just a short while," he reveals.

Balasubramanian explains why open source software (OSS) became the first choice for the project: "The primary technical requirements of the project were of scale and vendor neutrality at all levels. FOSS helped us achieve vendor-neutrality in many of our application components, which is very important for an initiative of national importance. The use of open standards has encouraged multi-vendor participation. This has driven costs down and improved the quality.

"The per-CPU licence costs that are normally associated with proprietary software have been largely eliminated. The benefits become more pronounced as the deployment footprint increases over thousands of CPU cores across multiple data centres.

"Access to the source code of the FOSS solutions has also helped us use them more effectively as we got a pretty good understanding of how it works, internally. The use of FOSS has also helped in its growing adoption by registrars and enrolment agencies, some of whom have strict guidelines around the use of proprietary software."

The strategy of adopting OSS solutions that are mainstream, have a strong community backing, and are actively developed and enhanced, works well most of the time, affirms Balasubramanian.